

Goserelin for prostate cancer (Zoladex[®], Zoladex LA[®])

Goserelin is a cancer drug. It is used to treat prostate cancer.

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What is goserelin (Zoladex[®], Zoladex LA[®])?

Goserelin is a hormonal therapy drug. It is also called Zoladex[®] or Zoladex LA[®]. It is used to treat [prostate cancer](#).

Goserelin can also be used to treat breast cancer. We have separate information about [goserelin for breast cancer](#).

It is best to read this information with our general information about [hormonal therapy](#) and the [type of cancer](#) you have.

Your cancer team will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

How goserelin works

Hormones are chemicals that our bodies make. They act as messengers and help control how cells and organs work. Hormonal therapy drugs change how hormones are made or work in the body.

Prostate cancer needs the hormone testosterone to grow. This is mainly made by the testicles.

Goserelin stops the testicles from making testosterone. This lowers testosterone levels and may shrink the prostate cancer or stop it growing.

Having goserelin

You have goserelin as a small pellet injected under the skin (subcutaneously) of your tummy (abdomen). The pellet releases the drug slowly as it dissolves.

During your treatment, you will meet someone from your cancer team, such as a:

- cancer doctor
- specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

During your course of treatment, you may also have regular blood tests.

If you are having Zoladex[®], you have it every 4 weeks. If you are having Zoladex LA[®], you have it every 12 weeks.

Your GP or practice nurse may give it to you at your GP surgery. Or a district nurse may be able to give it to you at home if you are not able to visit the surgery. They may use a local anaesthetic cream to numb the area before the injection.

If you are taking any medicines to thin your blood, tell your doctor, nurse or pharmacist. These medicines can make you bruise more easily.

When you start taking goserelin, you often have another [hormonal drug](#) with it. But this is only for a few weeks. Your cancer team will discuss your treatment plan with you.

Injection site

The skin in the area that was injected may become:

- painful
- swollen
- red, if you have white skin
- darker, if you have black or brown skin.

If this happens, let your doctor know. Painkillers may help.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

When you have this treatment, you will have regular appointments with a doctor, nurse or pharmacist. Always tell them about any side effects you have. They can give you:

- drugs to help control most side effects
- advice about managing side effects.

Most side effects can be managed. But sometimes side effects are harder to control. It is important not to stop taking hormonal therapy without telling your doctor. If side effects cannot be managed, your doctor may suggest a different type of hormonal therapy.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Sexual effects

It is common to lose your [sex drive](#) and have erection difficulties with hormonal therapy. Hormonal therapy may also make your penis shorter and your testicles smaller.

Your doctor can prescribe drugs and treatments to help with erection difficulties. These will not increase your sex drive. But they might work, even with a low sex drive.

Things can improve a few months after you stop taking the drug. But it depends on how long you take hormonal therapy, and what other treatments you have had. Some people continue to have problems after treatment ends.

Hot flushes and sweats

Hot flushes are a common side effect of this treatment. During a flush, your neck and face may feel warm. If you have white skin, your skin may look red. If you have black or brown skin, your skin colour may deepen. Flushes can last up to 5 minutes. You might also have sweats or feel anxious or irritable during a flush.

Here are some things you can do to reduce the effects of hot flushes:

- Wear natural fabrics, such as cotton.

- Wear layers so you can remove clothes as needed.
- Use cotton sheets and layers of bedding you can remove.
- Try cooling pads or pillows to keep you cool.
- Keep rooms cool or use a fan.
- Have cold drinks and avoid caffeine and alcohol.

You might have fewer hot flushes as your body adjusts to this treatment. You could try therapies to help you [cope with hot flushes](#), such as:

- [talking therapies](#) like cognitive behaviour therapy (CBT)
- controlled breathing exercises
- yoga
- acupuncture.

If they do not improve, talk to your doctor. Certain drugs can help to improve hot flushes.

Flushes usually stop a few months after treatment ends. But some people continue to have them.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Tumour flare

When you start taking this treatment your testosterone levels may go up for a very short time. This could cause any symptoms to get temporarily worse. This is sometimes called tumour flare. To prevent this, your doctor gives you another tablet to take before you start and for the first few weeks of treatment. Your doctor, nurse or pharmacist will explain more about this.

Mood changes

You may feel low or depressed, or have mood swings, during this treatment. Talking to family and friends about how you feel might help. If mood changes last for more than a few weeks, tell your doctor, nurse or pharmacist. They can talk to you about different ways to manage low mood or [depression](#).

Back pain, numbness and pins and needles

This treatment can cause back pain. It can also cause numbness or a tingling feeling, like pins and needles, in your legs.

Sometimes these symptoms are signs of a serious condition called [metastatic spinal cord compression or MSCC](#). This can happen when [advanced prostate cancer](#) that has spread to the bones of the spine presses on the spinal cord.

MSCC is not common, but it needs to be treated quickly. This is to stop permanent damage to the nerves.

If you have any of the symptoms below, get medical advice immediately. Contact the hospital on the 24-hour number. Or if you are unable to get in touch with anyone, go to the nearest Emergency Department (A&E) straight away:

- New pain that feels like a band around your chest or tummy.
- New pain that spreads down an arm or leg or into your lower back or buttocks.
- Pain that is severe, gets worse with movement or disturbs your sleep.
- Weakness in your arms or legs.
- Unsteadiness.
- Numbness or pins and needles in your legs.
- Bladder or bowel problems such as constipation, not passing much pee (urine) or poo (stool), or problems controlling your bladder and bowel (incontinence).

Bone thinning (osteoporosis)

Taking this treatment for several months or more can increase the risk of [bone thinning](#). This is called osteoporosis. It can make you more likely to get a broken bone (fracture).

You may have bone density scans to check your bone health before and during treatment. These scans can diagnose osteoporosis or osteopenia (low bone density) so that it can be monitored. If your scan shows osteoporosis, your doctor usually prescribes:

- drugs to protect your bones called [bisphosphonates](#)
- calcium and vitamin D supplements.

Regular exercise where you support your body weight, like walking, running and dancing, can help look after your bones. This is called [weight-bearing exercise](#). Other types of strength exercises are also helpful. If you have any bone thinning or the cancer is affecting your bones, get advice from your cancer team first.

Eating a healthy diet and not smoking also helps take care of your bones.

Raised blood sugar levels

This treatment can raise your blood sugar levels and increase the risk of diabetes.

Signs of raised blood sugar include:

- feeling thirsty
- needing to pass urine (pee) more often than usual
- feeling more tired than usual.

Tell your doctor or nurse if you have these symptoms. They can do tests to check your blood sugar level and give you advice.

If you already have [diabetes](#), your blood sugars may be higher than usual. You may need to check them more often, or make changes to your diabetic treatment. Your diabetic doctor or nurse will talk to you about how to manage this.

Tingling in fingers or toes

Goserelin can affect the nervous system. You may have pins and needles or feel tingling in your hands and feet. This is usually mild. If you notice these symptoms, contact the hospital on the 24-hour number as soon as possible.

Memory and concentration

You may notice [changes in your memory](#). You may also find it harder to concentrate.

To help you remember things, try using:

- reminder alarms or notes on your phone
- a diary, calendar or phone app
- a notebook you carry with you.

Skin changes

This treatment can [affect your skin](#). It may cause a rash, which may be itchy. Your doctor or nurse can tell you what to expect. If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

Always tell your doctor or nurse about any skin changes. They can give you advice and may prescribe creams or medicines to help. Changes to your skin usually improve when treatment ends.

Hair thinning

While you are having this treatment, the hair on your head might become thinner, drier and more brittle. This is usually mild.

You might also lose some:

- body hair, such as chest and pubic hair
- facial hair.

If you are worried about this, talk to your nurse about [hair care](#).

Weight gain

You may gain weight when you are having this treatment. [Eating healthily and being active](#) can help you keep to a healthy weight. Your doctor, nurse or dietitian can give you more advice.

Muscle or joint pain

You may get pain in your muscles or joints. If this happens, tell your doctor or nurse. They can give you advice and painkillers or other drugs to help. Being physically active and keeping to a healthy weight may help with the pain. It can keep your joints flexible and reduce stress on them. If the pain

does not get better, tell your doctor or nurse. They may talk to you about changing to a different hormonal drug.

Blood pressure changes

This treatment may cause low or high blood pressure. If you have ever had any problems with your blood pressure, tell your doctor or nurse. Let them know if you feel dizzy or have any headaches.

Effects on the heart

This treatment might increase certain risk factors for [heart problems](#). Tell your doctor if you have:

- had any problems with your heart
- high blood pressure
- high levels of cholesterol – a fatty substance in your blood.

They can monitor or treat these if needed. Talk to your doctor if you are worried about this.

Things you can do to take care of your heart include:

- not smoking
- maintaining a healthy weight
- being physically active.

Breast swelling or tenderness

This treatment may cause swelling and tenderness of your breast tissue. This is called gynaecomastia. To prevent this, some people have 1 or more low-dose radiotherapy treatments to the chest before treatment starts. Another type of hormonal drug might be used to treat the breast swelling. Your doctor can give you more advice.

Allergic reaction

This treatment might cause an allergic reaction. But this is not common. Signs of a reaction can include:

- a skin rash or itching
- feeling breathless or wheezy
- swelling of your face, mouth or throat.

If you feel unwell or have any of these signs, contact the hospital straight away on the 24-hour number. Do not take any more of this treatment until you have checked with them.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

Effects on the liver

This treatment may affect how your liver works. This is usually mild and goes back to normal after treatment ends. You will have blood tests to check how well your liver is working.

Other important information

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Contraception

Your doctor, nurse or pharmacist will advise you not to [make someone pregnant](#) while having this treatment. The drug may harm the developing baby. It is important to use effective contraception during treatment and for a while after it finishes.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

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Instructions on how you can make a PDF version of this page using the print function on your browser.

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help